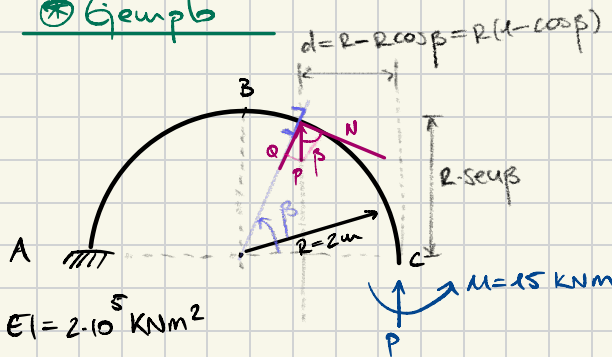


Arcos

Ejemplo



a) Leyes de Esfuerzos

b) Mov. de B.

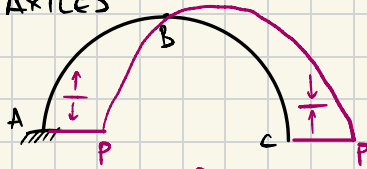
$$\rightarrow / \curvearrowright N(\beta) = P \cdot \cos \beta$$

$$\nearrow / \uparrow Q(\beta) = P \cdot \sin \beta$$

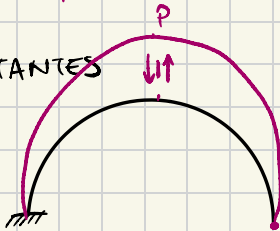
$$\nearrow / \curvearrowright M(\beta) = M + P \cdot R(1 - \cos \beta)$$

a) Leyes

AXILES



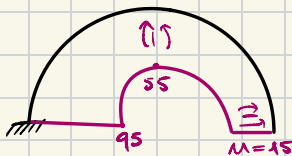
CORTANTES



$$\uparrow V_c = \int_0^\pi \frac{M(\theta)}{EI} \cdot ds \cdot R(1 - \cos \beta) = \int_0^\pi \frac{15 + 40(1 - \cos \beta)}{2EI} \cdot 2d\beta \cdot 2(1 - \cos \beta)$$

$$\uparrow V_c = 4,71 \cdot 10^{-3} \text{ m}$$

FLECTORES



$$\vec{M}_c = \int_0^\pi \frac{M(\theta)}{EI} \cdot ds \cdot R \sin \beta = \int_0^\pi \frac{15 + 40(1 - \cos \beta)}{2EI} \cdot 2d\beta \cdot 2 \sin \beta$$

$$\vec{M}_c = 2,2 \cdot 10^{-3} \text{ m}$$

Mov. en B: $\pi/2$
 El resto igual.

$$M(0) = M$$

$$M(\pi/2) = M + PR$$

$$M(\pi) = M + 2PR$$

⊗ PROBLEMA 3.9

DATOS

$$\Delta T = 40^{\circ}\text{C}$$

$$EI = 6ET \text{ N}\cdot\text{m}^2$$

$$\alpha = 10^{-6} \text{ }^{\circ}\text{C}^{-1}$$

a) Factores en A-B-C

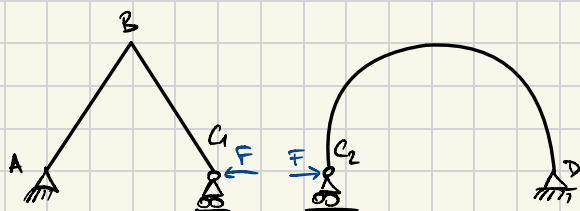
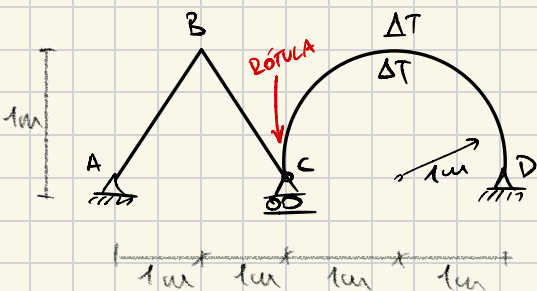
b) Esfuerzos en C-D

c) Reacciones

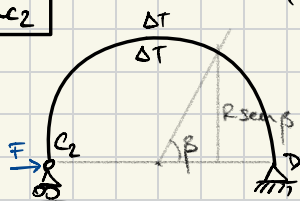
d) $\vec{u}_C$, e) $\vec{v}_B$

$$GH = 1 \text{ (rotula)}$$

$$= 5 \text{ inc.} - 3 \text{ ec.} - 1 \text{ rot.} = 1.$$

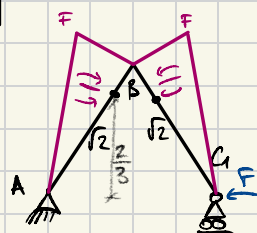


$$\vec{u}_{C2}$$



$$M(\beta) = F \cdot 2 \text{ sen } \beta$$

$$u_{C1}$$



$$\vec{u}_{C1} = \vec{0} \cdot d + \frac{1}{EI} \cdot 2 \left[\frac{1}{2} \sqrt{2} \cdot F \cdot \frac{2}{3} \right]$$

$\rho(d=0)$

$$\vec{u}_{C1} = \frac{2\sqrt{2} F}{3EI}$$

$$\vec{u}_{C2} = \int_0^{\pi} \frac{-M(\beta)}{EI} ds \cdot 2 \text{ sen } \beta + \alpha \cdot \Delta T \cdot 2R = \dots$$

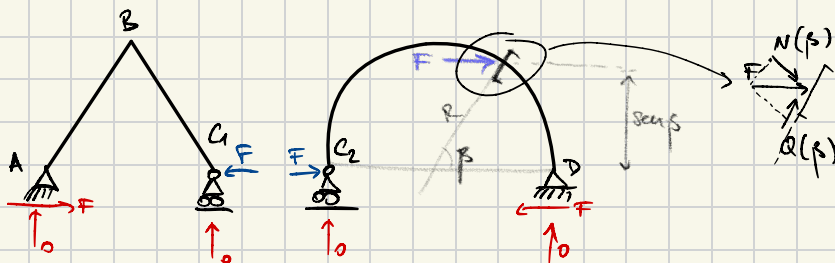
por F ⊗

R · dβ

(mov. horiz)

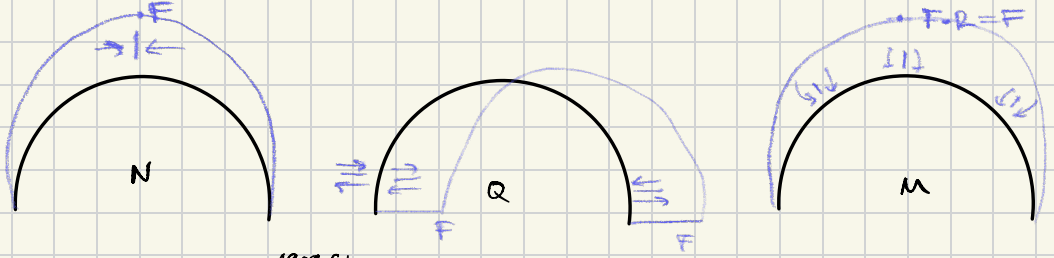
$$\vec{u}_{C2} = -\frac{F\pi}{2EI} + 80 \cdot 10^{-6}$$

Igualemos: $u_{C1} = u_{C2} \rightarrow F = 1909,61 \text{ N}$



$$N(\beta) = F \cdot \text{sen } \beta$$

$$Q(\beta) = F \cdot \text{cos } \beta$$



$$\overline{U}_C = \overline{U}_{C1} = \frac{2\sqrt{2} F}{3EI} = \boxed{3 \cdot 10^{-5} \text{ m} = \overline{U}_C}$$

1909,61
667

e) ¿ $\hat{V}_B$?

$$\hat{V}_B = \hat{Q}_A \cdot 1 - \frac{1}{EI} \left[\frac{1}{2} \sqrt{2} \cdot F \cdot \frac{1}{3} \cdot 1 \right] = 1,5 \cdot 10^{-5} \text{ m}$$

$$\hat{V}_{C1} = 0 = \hat{Q}_A \cdot 2\text{m} - \frac{1}{EI} \left[\frac{1}{2} \sqrt{2} F \left(\frac{4}{3} + \frac{2}{3} \right) \right] \rightarrow \hat{Q}_A = \frac{\sqrt{2} \cdot F}{2EI}$$

$1 + \frac{1}{3}$

⊕ PROBLEMA 3.6

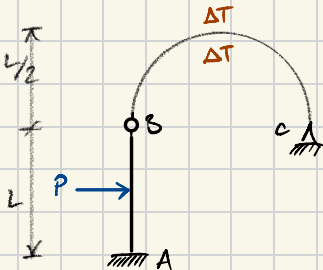
a) GH

b) REACCIONES

c) U_B

$$a) GH = \underbrace{L_F}_S - \underbrace{g_{KE}}_S - 2\text{rot} = 1.$$

b) Libramos el mov. horiz. en c porque si separamos x la rotula, se convierte en un mecanismo.



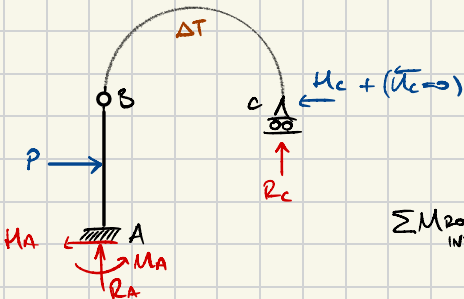
$$\sum F_H = 0 \quad P - U_C - H_A = 0 \quad \underline{H_A = P - U_C}$$

$$\sum M_{\text{rot}, \text{DER.}} = 0 \rightarrow R_C \cdot L = 0 \quad \underline{R_C = 0}$$

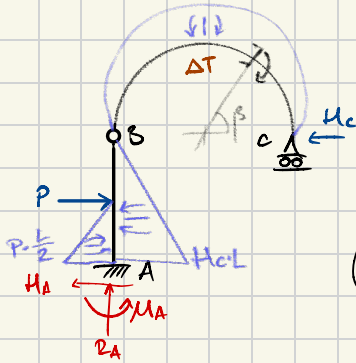
$$\sum F_V = 0 \rightarrow R_A + 0 = 0 \quad \underline{R_A = 0}$$

$$\sum M_{\text{rot}, \text{INF}} = 0 \rightarrow M_A - (P - U_C) \cdot L + P \frac{L}{2} = 0$$

$$\underline{M_A = \left(\frac{P}{2} - U_C \right) L}$$



$$\overleftarrow{U}_c = 0 = \overleftarrow{U}_B - \alpha \cdot \Delta T \cdot L + \int_0^{\pi} \left(\frac{H_c \cdot 2 \sin \beta}{EI} \cdot 2 d\beta \cdot 2 \sin \beta \right)$$



$$\rightarrow H_c = \frac{\frac{5P}{48} + \alpha \cdot EI \cdot \Delta T}{\frac{1}{3} + \frac{\pi}{16}}$$

$$\frac{H_c L^3 \pi}{16 EI} \rightarrow \text{Con } H_c \rightarrow \begin{cases} H_A \\ R_A \\ M_A \\ H_c \\ R_c \end{cases}$$

$$\overleftarrow{U}_B = \frac{1}{EI} \left[\frac{L}{2} H_c \cdot L \frac{2L}{3} - \frac{1}{2} \frac{L}{2} \frac{PL}{2} \left(\frac{L}{2} + \frac{2}{3} \cdot \frac{L}{2} \right) \right]$$

$$\text{Con } H_c: \overleftarrow{U}_B = \frac{1}{EI} \left[\frac{H_c}{3} - \frac{5P}{48} \right]$$